CISC 121 homework 1

Q1) CyclicRotation

```
function solution(A, K):
    if A is empty:
        return A

    result = copy of A

    repeat K times:
        last = remove the last element of result
        put last at the front of result

    return result
```

Q2) OddOccurencesInArray

```
function solution(A):
    counts = empty dictionary

    for each number in A:
        if number already in counts:
            add 1 to its count
        else:
            set its count to 1

    for each number in counts:
        if its count is odd:
            return that number
```

Q3) Frog Jmp

```
function solution (X, Y, D)
    distance = Y - X
    jumps = distance div D    //whole jumps

    if distance mod D is not 0:
        jumps = jumps + 1
        //one more jump for the leftover

    return jumps
```

Q4) PermMissingElem

```
function solution(A):
    N = length of A

    // Sum of every number from 1
to n + 1
    expectedSum = 0
    for number from 1 to N + 1:
        expectedSum = expectedSum + number

    // sum of the numbers actually in
the array
    actualSum = 0
    for each number in A:
        actualSum = actualSum + number

    return expectedSum = actualSum
```

Q5) Tape Equillibrium

```
function solution (A):
    N = length of A

    // add up the whole array once
    total = 0
    for each number in A:
        total = total + number

    leftSum = 0
    minDifference = very large number

    // P is the split point, from 1 to n - 1
    for P from 1 to n - 1
        leftSum = leftSum + A[P - 1]
        rightSum = total - leftsum

        difference = absolute value of (leftSum - rightSum)

        if difference < minDifference
            minDifference = difference

    return minDifference
```